PCM Tune-1: bounded ternary hyperparameter search

20% sampling, NPL-default noise, 256-by-256 images

Preparation report, version 1.0

Status: preparation complete. No tune-1 GPU search or test45 evaluation has run.

1 Executive summary

We prepare an experiment with two independent PnP algorithms: PnP-ADMM-DRUNet and GS-PnP-GS-DRUNet. *Test set 450* remains the full original test set (90 images in each of five categories). *Test set 45* contains nine randomly drawn held-out test images per category, sampled with a fixed random seed. The 45 image IDs and their original measurement hashes are frozen.

We select **five validation calibration images per category**. Each image produces one independent tune-1 search. There are therefore five category experiments containing **25 independent searches for each PnP algorithm**, or 50 searches altogether. Each resulting configuration is applied unchanged to the same nine held-out test images of its category. Parameters are selected only from noisy acquired measurement cross-validation; test PSNR, SSIM and image ground truth are never used for tuning.

2 Tuning algorithm

For an acquisition with Hadamard model $y = Ax + \varepsilon$, let Λ_k be one of four pre-frozen held-out sets of acquired fine-scale Hadamard rows, and let Θ_k be its complement within the acquired rows. The first 4,096 coarse patterns always stay in the fit set. The original 9,011 other acquired patterns are divided among four held-out folds. For candidate θ, calculate

$$L(\theta) = \frac{1}{4}\sum_{k=1}^{4} \frac{\|A_{\Lambda_k}\widehat{x}_\theta(A_{\Theta_k}, y_{\Theta_k}) - y_{\Lambda_k}\|_2^2}{|\Lambda_k|}.$$

Each image has its own loss and independently selected configuration. Every candidate is evaluated using the *same* four measurements splits for that calibration image.

Log-coordinate ternary search

The previous 15- and 18-candidate grids were Cartesian products of specific parameter values. The new strategy instead specifies continuous positive ranges and shrinks them numerically. For a single parameter, let $u = \log_{10} p$ and start with $[a, b] = [\log_{10} p_{\min}, \log_{10} p_{\max}]$. At each step, evaluate two interior points,

$$u_1 = a + (b - a)/3, \qquad u_2 = b - (b - a)/3.$$

Under the assumption that the one-dimensional loss slice is *unimodal*, retain $[a, u_2]$ if $L(u_1) \le L(u_2)$, otherwise retain $[u_1, b]$.

With several parameters, hold the other coordinates fixed, perform three ternary contractions for one coordinate, then move to the next. Use two passes, reversing the coordinate order in the second pass. Keep the best *measured, numerically valid* configuration. One-dimensional unimodality does **not** imply a globally unimodal two- or three-dimensional objective. Discontinuous solver stopping and backtracking can violate this assumption in practice.

Algorithm	Parameter	Frozen bounds
PnP-ADMM	ρ	0.03 to 4.0
PnP-ADMM	Denoiser σ	0.015 to 0.18
GS-PnP	λ	0.005 to 0.40
GS-PnP	Denoiser σ	0.015 to 0.15
GS-PnP	Initial τ_0	3.0 to 75

Use the original pre-trained denoisers, normalised PCM fidelity and solver stopping rules. Every candidate is scored on all four SSDU folds. Numerical failures invalidate a candidate; they must be recorded rather than assigned a fictitious finite score.

3 Step-by-step plan

1. **Freeze sample selection.** Select five validation and nine test images per category, independently of reconstruction metrics. Save RNG seeds, image IDs and measurement/mask SHA256 fingerprints. **Completed.**
2. **Verify search code.** Check interval shrinking on synthetic unimodal objectives, reproducibility, output hashes and test-access guards. **Completed in the CPU preparation tests.**
3. **Perform 50 tune-1 searches.** Run 25 ADMM and 25 GS-PnP independent searches on the preselected noisy validation data. Save all probed parameters, four fold losses, solver diagnostics, and failed configurations.
4. **Freeze all parameters before test evaluation.** Test-set access is blocked until every one of the 50 independent selection records has been frozen.
5. **Apply parameters to test45.** For each calibration, reconstruct each of the nine test images in the same category, using the full 20% acquired measurement set. This produces 225 full reconstructions per PnP family.
6. **Compare against established methods.** Reuse archived 450-image baseline metrics after checking case and acquisition hashes. Regenerate any missing historical image arrays under the original fixed settings, then verify their image metrics.
7. **Aggregate correctly.** For each tune-1 calibration, calculate the mean and sample SD over nine test images. To report one test45 number for tune-1, average five calibration results for each *unique* test image first, then average across the 45 unique test images. The repeated calibrations are not 225 independent test-image samples.
8. **Prepare deliverables.** Save individual reconstructions and scores, paired comparisons for PSNR, SSIM, HaarPSI and MSE, per-category tables and plots, and 45 visual comparison pages showing all historical and new tune-1 results.

4 Experiment running

Frozen image IDs (per category):

Category	Five validation calibrations	Nine test45 images
CIGS	0008, 0530, 0589, 0733, 0768	0068, 0095, 0254, 0292, 0304, 0507, 0527, 0710, 0935
Organic	0158, 0530, 0589, 0733, 0768	0014, 0055, 0116, 0279, 0470, 0518, 0862, 0935, 0946
Perovskite	0008, 0045, 0733, 0768, 0917	0021, 0187, 0292, 0301, 0376, 0610, 0727, 0767, 0786
Reference	0008, 0045, 0158, 0396, 0768	0021, 0055, 0129, 0199, 0260, 0357, 0702, 0793, 0946
Silicone	0008, 0158, 0283, 0589, 0733	0098, 0116, 0188, 0260, 0507, 0606, 0711, 0767, 0933

Both PnP solvers use the same 20% Hadamard sampling rate and fixed NPL-default noise already archived for validation and test. No noise or masks will be regenerated for parameter selection. The planned GPUs are tuni RTX 4070 (ADMM) and RTX 4090 (GS-PnP). The prepared runner isolates each GPU and saves scores after each calibration candidate, allowing safe resumption.

Two parameter passes with three ternary contractions per coordinate require at most 25 distinct candidate evaluations per ADMM calibration and 37 per GS calibration. At four folds per candidate, upper budgets are $25 \times 25 \times 4 = 2{,}500$ ADMM solver calls and $25 \times 37 \times 4 = 3{,}700$ GS solver calls. A solver call can take up to 100 iterations and GS-PnP may require additional network calls during backtracking. These are *planned budgets*, not completed computations.

5 Results and deliverables

No new tune-1 parameter results have been generated. The 25 calibration acquisitions and 45 test cases are frozen. The report will be updated after all 50 validation-only parameter selections and the test45 reconstructions complete.

The preparation archive includes the frozen test45 and validation sample lists, parameter bounds, tested pure ternary-search code, validation-only GPU tuning runner, test45 reconstruction runner, historical baseline

replay scripts, metric aggregator and qualitative gallery generator. Full numerical results will only be inserted after all searches and reconstructions finish.

6 Conclusion

The experiment is prepared, not yet executed. Ternary search assumes one-dimensional unimodality along each coordinate; the joint PnP loss need not have one optimum. Test metrics will not be used to choose parameters.

Primary sources. Yaman et al. (2020), *Self-Supervised Learning of Physics-Guided Reconstruction Neural Networks without Fully-Sampled Reference Data*, section on measured-data holdout: https://doi.org/10.1002/mrm.28378.

Berkeley Statistics 243, *Univariate Function Optimisation*, section on unimodal search and interval bracketing: https://stat243.berkeley.edu/fall-2024/units/unit11-optim.html.

Hurault, Leclaire and Papadakis (2022), *Gradient Step Denoiser for Convergent Plug-and-Play*, GS-PnP update and backtracking: https://openreview.net/pdf?id=fPhKeld3Okz.